

Companion LXVII: Cross-Covariance of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

The cross-covariance of persistence diagrams quantifies the correlated topological structure between different realizations, smoothing scales, or redshift slices of a cosmological field. In weak-lensing analyses, this covariance captures how connected components, loops, and void-like structures co-vary across the gravitational field. In the standard cosmological model, cross-covariance reflects the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the correlation structure of topological features in a characteristic way. This Companion develops a continuous description of cross-covariance in RDCC, deriving how the relaxation scale influences correlated topology, persistence-lifetime coupling, and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive probe of the RDCC gravitational field through correlated topological structure.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. While ensemble variance measures the spread of topological features across realizations, the cross-covariance quantifies how topological structures co-vary between different realizations, smoothing scales, or redshift slices.

In the standard cosmological model, cross-covariance reflects the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the correlation structure of topological features in a scale-dependent manner, producing a distinctive signature in cross-covariance analyses.

2 Cross-Covariance of Persistence Diagrams

Given two ensembles of persistence diagrams $\{D_i\}$ and $\{E_i\}$ with Fréchet means D^* and E^* , the cross-covariance is defined as

$$\text{Cov}(D, E) = \frac{1}{N} \sum_{i=1}^N W_p(D_i, D^*) W_p(E_i, E^*), \quad (1)$$

where W_p is the p -Wasserstein distance.

This covariance measures the degree of correlated topological structure.

3 RDCC Modification of Persistence Diagrams

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (2)$$

This modifies the birth and death thresholds of topological features:

$$b_{\text{RDCC}} = b [1 - \chi_b (\ell/\ell_{\text{rel}})^{\alpha}], \quad (3)$$

$$d_{\text{RDCC}} = d [1 - \chi_d (\ell/\ell_{\text{rel}})^{\alpha}]. \quad (4)$$

The RDCC ensembles become

$$\{D_{i,\text{RDCC}}\}, \quad \{E_{i,\text{RDCC}}\}. \quad (5)$$

4 Cross-Covariance in RDCC

The RDCC cross-covariance is

$$\text{Cov}_{\text{RDCC}}(D, E) = \frac{1}{N} \sum_{i=1}^N W_p(D_{i,\text{RDCC}}, D_{\text{RDCC}}^*) W_p(E_{i,\text{RDCC}}, E_{\text{RDCC}}^*). \quad (6)$$

To first order in the relaxation parameter,

$$\text{Cov}_{\text{RDCC}}(D, E) = \text{Cov}(D, E) \left[1 - \chi_{\text{Cov}} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (7)$$

This modification reflects the relaxation-driven change in correlated topology.

5 Correlated Topological Signatures of RDCC

The relaxation mechanism enhances the coherence of large-scale modes. Cross-covariance therefore exhibits:

- enhanced correlation of large-scale topological features,
- suppressed correlation of small-scale features,
- a characteristic turnover near ℓ_{rel} ,
- modified redshift evolution of topological coupling,
- altered cross-scale persistence-lifetime correlations.

These signatures provide a direct probe of the RDCC gravitational field through correlated topological structure.

6 Conclusion

The cross-covariance of persistence diagrams in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the correlation structure of topological features in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of correlated topology, nonlinear structure, and the interplay between light and gravity in RDCC.

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